



RExcharge >>>>

ESUM Theme 2

Case Study Guide

Each team shall propose **one working prototype or proof-of-concept**.

The prototype shall be capable of:

1. Outputting the observation result
2. Able to identify charger serial number & brand/model
3. Identifying the fault type
4. Routing the issue to the correct recipient (customer or after-sales team)

A simple identifier (e.g., a number) may be used to represent the after-sales team. The output message shall be directed accordingly:

1. If the customer is the appropriate recipient, the result shall be displayed in the interface
2. If the after-sales team is the recipient, the message shall be routed to the assigned team identifier within the prototype system

EV DISTRIBUTION BOARD



Observation :

- Missing MCB / RCCB
- Wrong Component /Specs

FAULT Type :

- Protection Issue

Action:

- Contact after-sales team to repair or replace missing or incorrectly specified breakers

CHARGER



Observation :

- Red Light
- Blinking Red Light
- No Light

FAULT Type :

- Charger Issue (Red Light, No Light, Blinking Red Light)
- Supply Issue (No Light, Blinking Red Light)
- Installation Issue (Blinking Red Light)
- Manual Error (Blinking Red Light)

Action :

- No Light: Advised customer to check whether the breaker in the EVDB has tripped.
- Red Light: Screenshot the app (if any) & send to after sales team
- Blinking Red Light: Refer to Error Logs Slide PG10

ISOLATOR



Observation

- Isolator OFF (Open Circuit)

FAULT Type :

- Power Cut

Action:

- Advised customer to turn ON Isolator
- Advised customer to check whether the breaker in the EVDB has tripped.

Observation

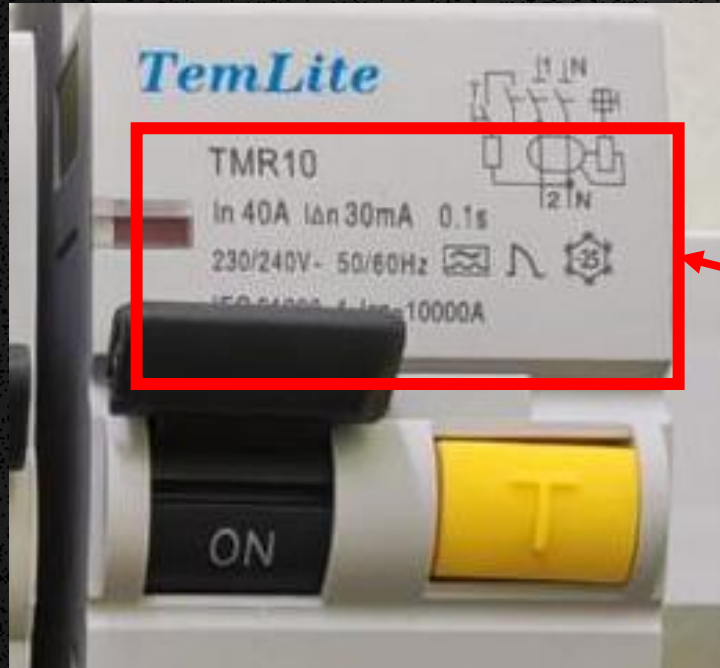
Fault Type

Action Steps

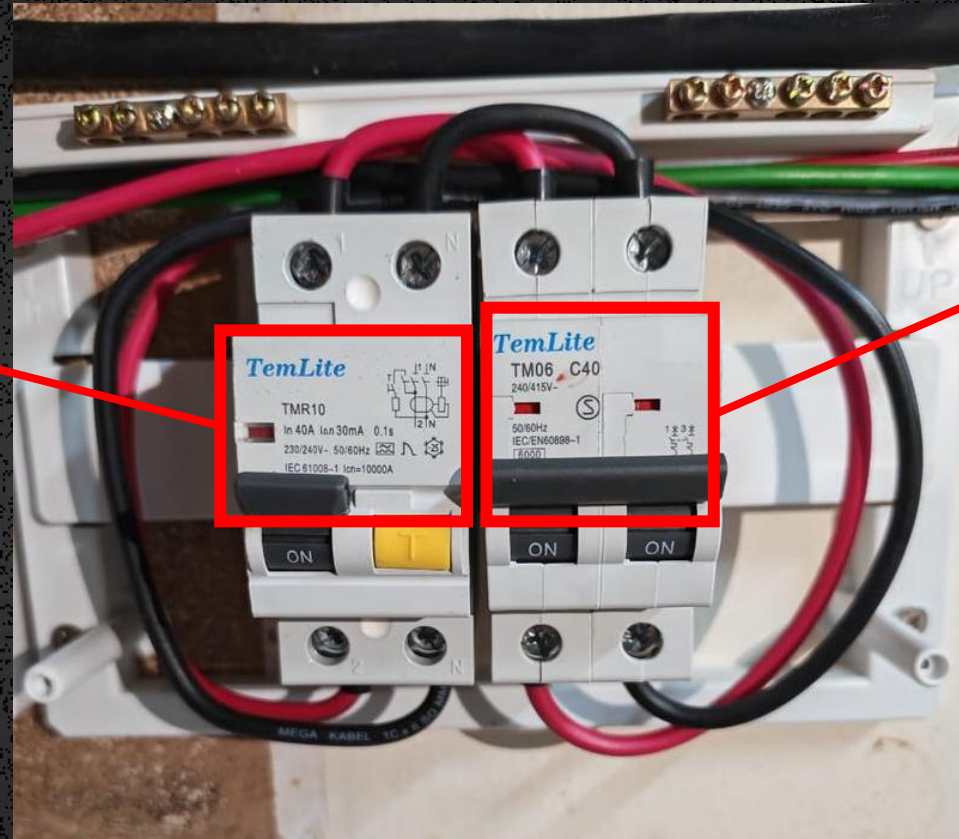


RExcharge >>>> ***EVDB as Input***

EVDB : Single Phase



1 Phase : RCCB 40A Type A 2P



1 Phase EVDB
Component / Safety
breaker

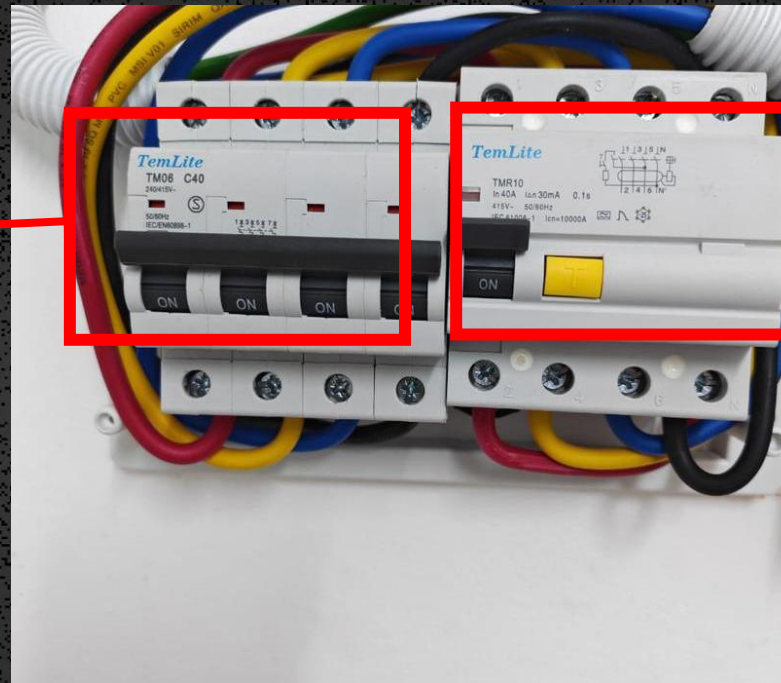


1 Phase : MCB 40A 2P

EVDB : Three Phase



3 Phase EVDB: MCB 40A 4P



3 Phase EVDB
Component / Safety
breaker



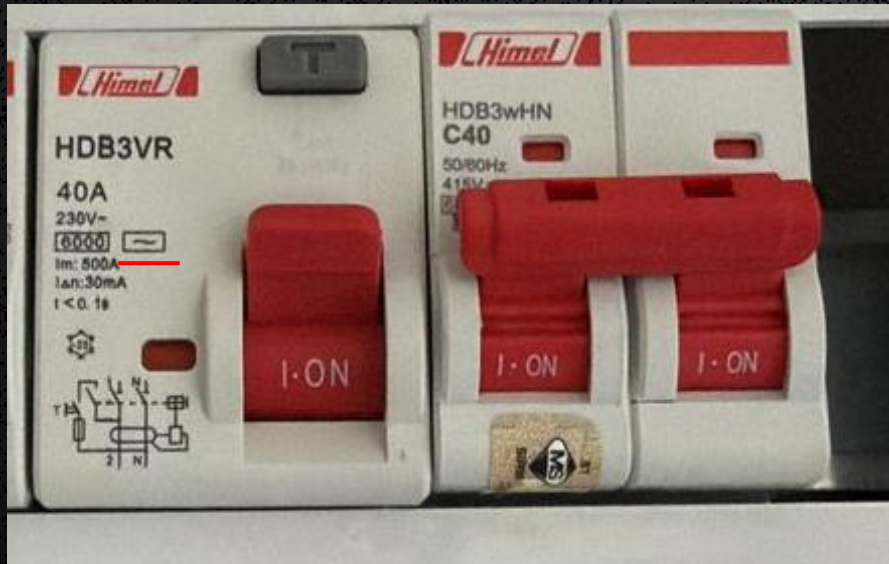
3 Phase : RCCB 40A Type A 4P

Example of Type RCCB Sinusoidal wave

AC sinusoidal wave



TYPE AC SINUSOIDAL



AC sinusoidal wave



Pulsating DC

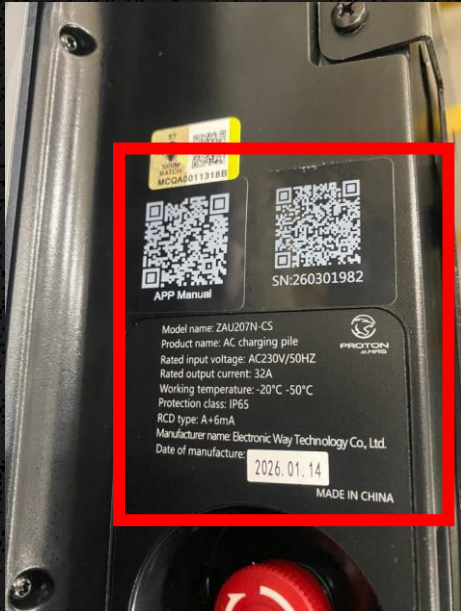
TYPE A SINUSOIDAL





RExcharge >>>> *Charger as Input*

CHARGER :



Charger Serial Number
and Charger Brand/Model

Charger SN : 260301982
Brand : Proton eMAS



Charger Indicator RED

Observation :

- Red Light/Blinking Red Light
- No Light

FAULT Type :

- Charger Issue (Red Light, No Light)
- Supply Issue (No Light)

Action :

- No Light: Advised customer to check whether the breaker in the EVDB has tripped.
- Red Light: Screenshot the app (if any) & send to after sales team
- Blinking Red Light: Refer to Error Logs Slide PG10

Error Logs:

Troubleshooting:

Note: Below faults should be tested and diagnosed by professional electricians. We take no responsibility for any damage or safety hazards caused by operation by non-professional person.

Indicator	Issue	Error Cause	Operation Taken
Red Light Flashes 6 times	Ground Fault	- Wrong or <u>bad connection</u> of wiring - Abnormal reading on Neutral to Earth	- Check on the condition of termination - Check on the reading especially voltage of Neutral to Earth
Red Light Flashes 7 times	Emergency Stop Button Error	- <u>Accidentally press</u> on emergency stop button	- Release the Emergency Stop button
Red Light Flashes 8 times	Short Circuit	- Short circuit happen	- Check voltage, current reading on charger - Check on cable condition to exclude cable burnt happen
Red Light Flashes 9 times	Over-temperature	- Temperature inside charger <u>exceeding normal range</u> - The charger is operating continuously in a long period	- Shut down the charger for a while, and restart the charger

Fault Type: Installation Issue
Action: Contact after sales team

Fault Type: Manual Error
Action: Advise Customer to release Emergency Stop Button

Fault Type: Charger Issue
Action: Contact after sales team

Fault Type: Charger Issue
Action: Advise Customer to shut down the charger a while & restart the charger



RExcharge >>>>

Isolator as Input



Isolator



ON & OFF



Isolator Switch ON

Isolator Switch OFF

Observation

- Isolator OFF (Open Circuit)

FAULT Type :

- Power Cut

Action:

- Advised customer to turn ON Isolator
- Advised customer to check whether the breaker in the EVDB has tripped.